Machine Learning. Lecture 5: PCA

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Outline

1. Support Vector Machine (SVM)
2. Dimensionality reduction and PCA
3. Validation strategies

Support Vector Machine

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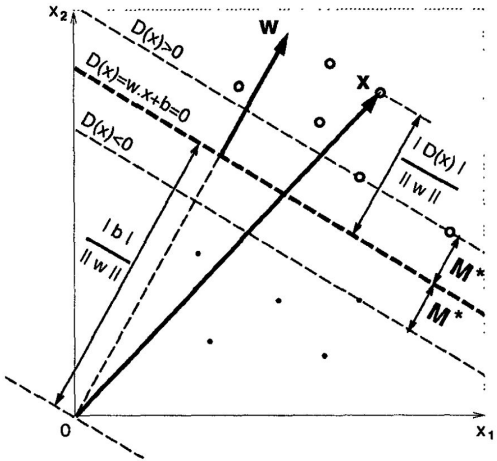
01

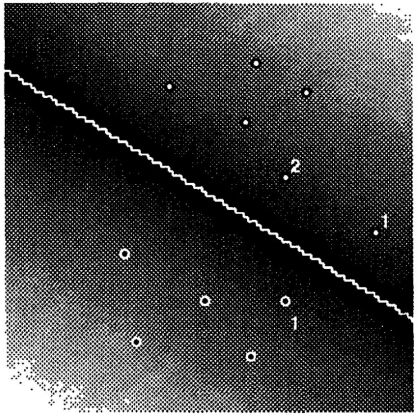
Support Vector Machine



1. History
2. Motivation
3. Solution for separable design
4. Inseparable design, soft margin
5. Kernels
    a. Kernel definition (Hilbert spaces, inner product, positive semidefiniteness)
    b. Kernels properties (addition, infinite sums)
    c. Types of kernels (poly, exponential, gaussian)
6. Current state

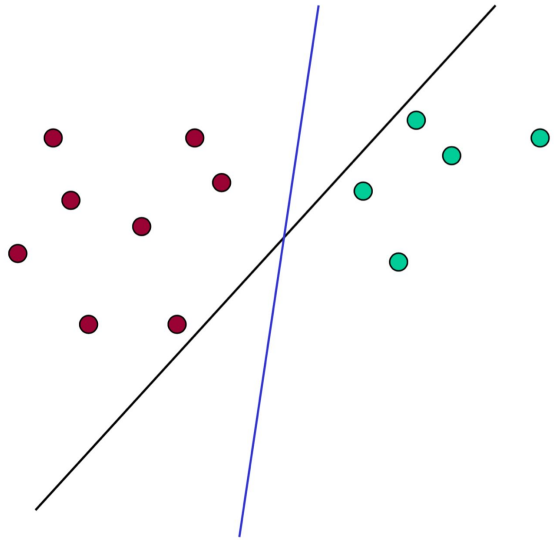
History





1963: SVM introduced by Soviet mathematicians Vladimir Vapnik and Alexey Chervonenkis
1992: kernel trick (Vapnik, Boser, Guyon)
1995: soft margin (Vapnik, Cortes)

Motivation



Linear separable case

Many separating hyperplanes exist

Maximize width

Margin

$$y \in \{1, -1\}$$

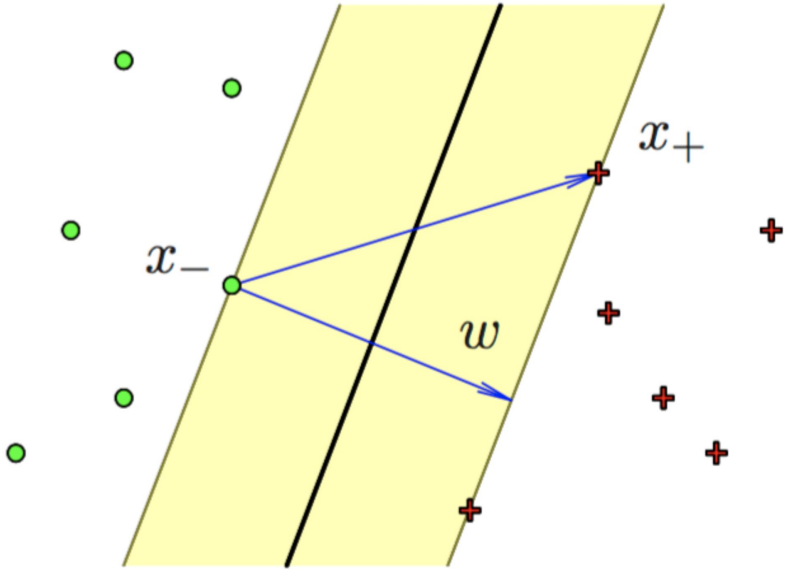
$$y_i = 1 : w^T x_i - c > 0$$

$$y_i = -1 : w^T x_i - c < 0$$

$$c_+(w) = \min_{y_i=1}(w^T x_i)$$

$$c_-(w) = \max_{y_i=-1}(w^T x_i)$$

$$\rho(w) = \frac{c_+(w) - c_-(w)}{2}$$



$$\rho\left(\frac{w_0}{||w_0||}\right) = \frac{1}{||w_0||}$$

Optimization problem

$$y_i = 1 : w^T x_i - c > 0$$
$$y_i = -1 : w^T x_i - c < 0$$
$$M_i = y_i \cdot (w^T x_i - c)$$

$$\rho(w) = \frac{1}{||w||} \to \max_{w,c}$$
$$s.t.\ y_i(w^T x_i - c) \geq 1$$

Convex problem!

$$L(w, c, \alpha) = \frac{1}{2} w^T w - \sum_i \alpha_i (y_i (w^T x_i - c) - 1)$$

Many of them are zeros

Hinge loss

$$L(w, c, \alpha) = \frac{1}{2} w^T w - \sum_i \alpha_i (y_i (w^T x_i - c) - 1)$$

$$L^{\text{hinge}} = (1 - M)_+$$

$$L(w, c, \alpha) = \frac{1}{2} ||w||_2^2 + \sum_i \alpha_i L_i^{\text{hinge}}$$

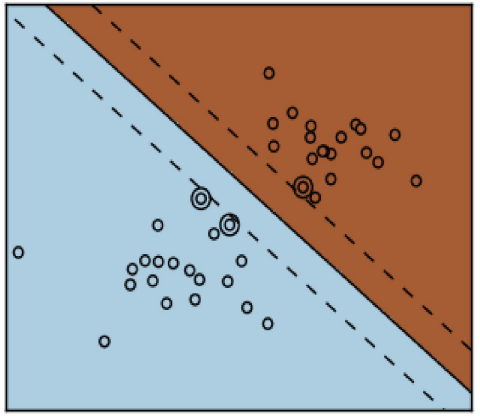
Inseparable case

Let our model mistake, but penalize that mistakes

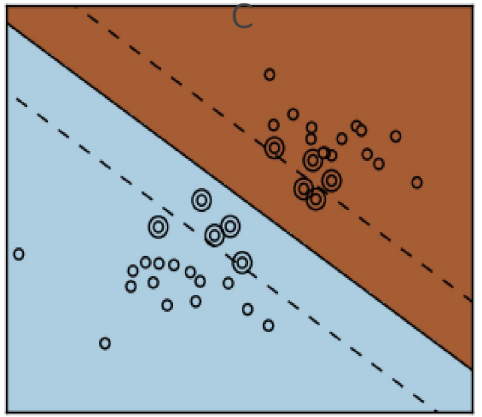
Implemented via margin slack variables

$$\begin{cases} \frac{1}{2}\langle w, w\rangle + C\sum_{i=1}^{\ell}\xi_i \to \min\limits_{w,w_0,\xi}; \\ y_i\big(\langle w, x_i\rangle - w_0\big) \geqslant 1 - \xi_i, \quad i = 1, \ldots, \ell; \\ \xi_i \geqslant 0, \quad i = 1, \ldots, \ell. \end{cases}$$

Big C



Small C



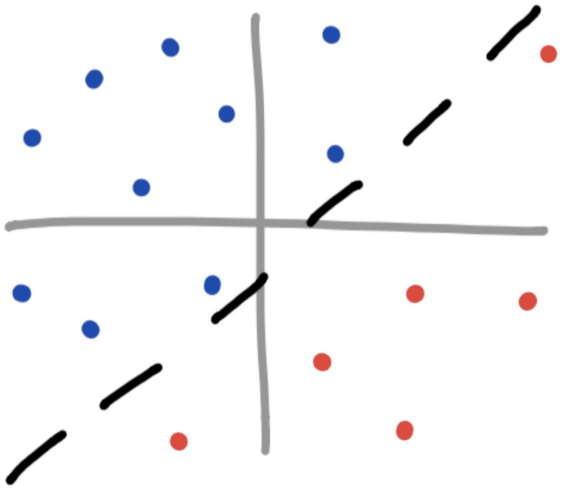
Kernel trick

$$y_i = 1 : w^T x_i - c > 0$$
$$y_i = -1 : w^T x_i - c < 0$$

$$\begin{matrix} x \mapsto \phi(x) \\ w \mapsto \phi(w) \end{matrix} \implies < w, x > \mapsto < \phi(w), \phi(x) >$$

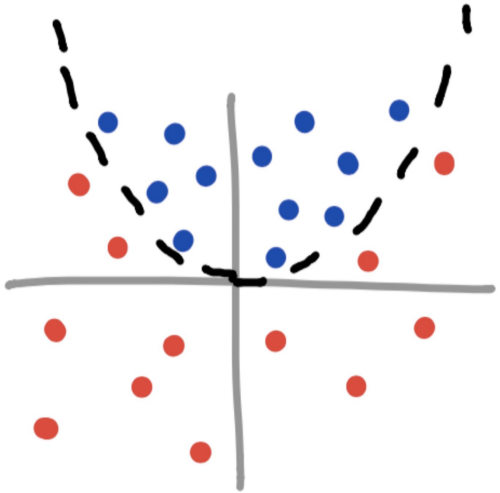
$$K(w, x) = < \phi(w), \phi(x) >$$

Kernel types



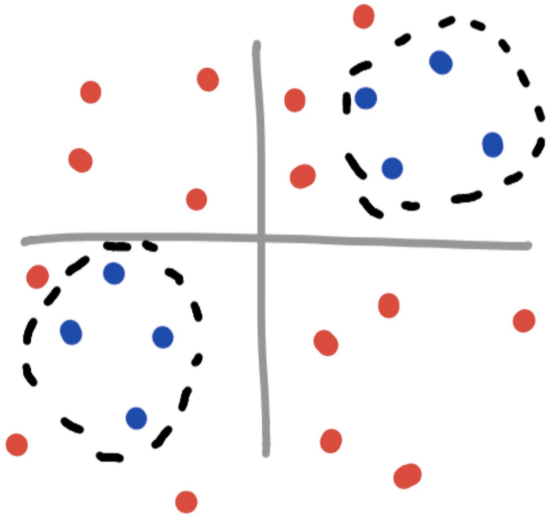
$K(w,x) =< \mathrm{w}, x >$

Linear



$K(w,x) = (\gamma < w,x > +r)^d$

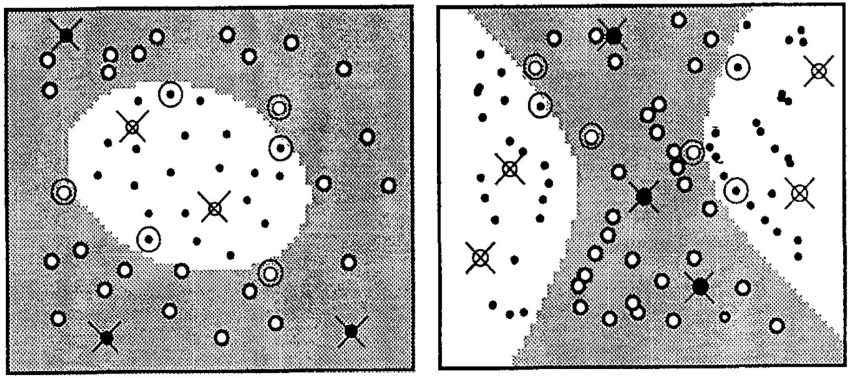
Polynomial



$K(w,x) = e^{-\gamma\|w-x\|^2}$

Gaussian radial basis function

Current state





Principal Component Analysis

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02

Principal Component Analysis

$$x_1, \ldots, x_n \rightarrow g_1, \ldots, g_k, k \leq n$$

$$U : UU^T = I, G = XU$$

$$\hat{X} = GU^T \approx X$$

$$||GU^T - X|| \rightarrow \min_{G,U} \; s.t. \; rank(G) \leq k$$

Singular value decomposition

$$||GU^T - X||_2 \leftarrow \min_{G,U} s.t. rank(G) \leq k$$

$$X = V\Sigma U^T : ||GU^T - V\Sigma U^T||_2 = ||G - V\Sigma||_2$$

$$G = V\Sigma' : ||V\Sigma' - V\Sigma||_2 = ||\Sigma' - \Sigma||_2$$

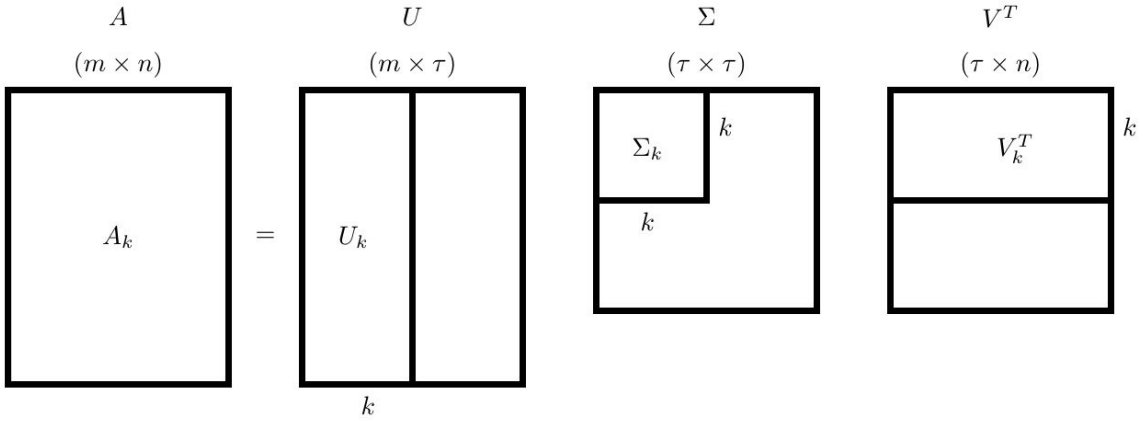
$$||A||_2 = \sigma_{max}(A) : ||\Sigma' - \Sigma||_2 = \sigma_k(\Sigma) = \sigma_k(X)$$

Eckart–Young–Mirsky theorem

Singular value decomposition

$$||GU^T - X|| \to \min_{G,U} \; s.t.\; rank(G) \le k$$

$$X = V\Sigma U^T \qquad \sigma_k(\Sigma) = \sigma_k(X)$$



Eckart–Young–Mirsky theorem

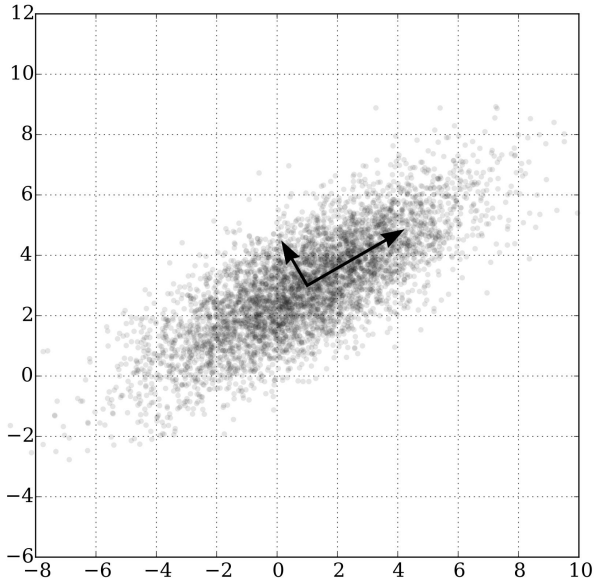
Another approach

Residual variance maximization

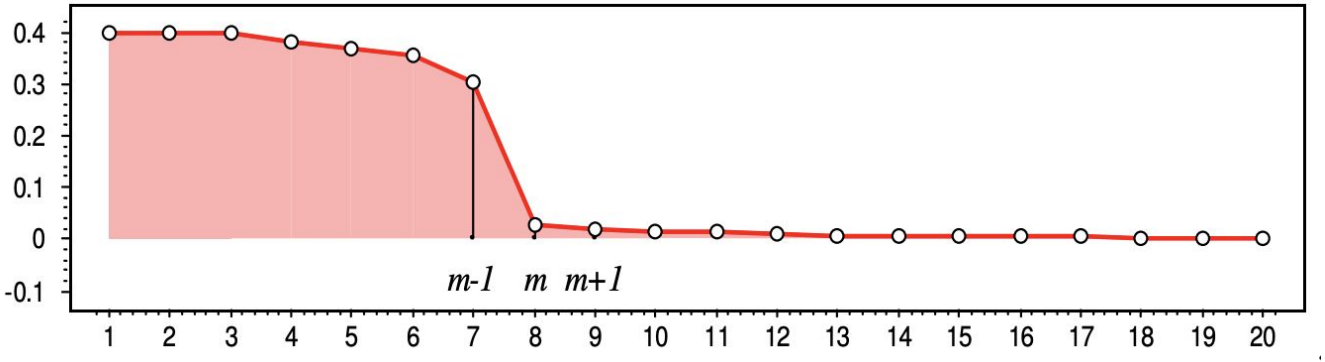
Take new basis vectors greedy

Same result for G and U

Always normalize data before PCA!!!



Dimensionality reduction



Get rid of low-variance components

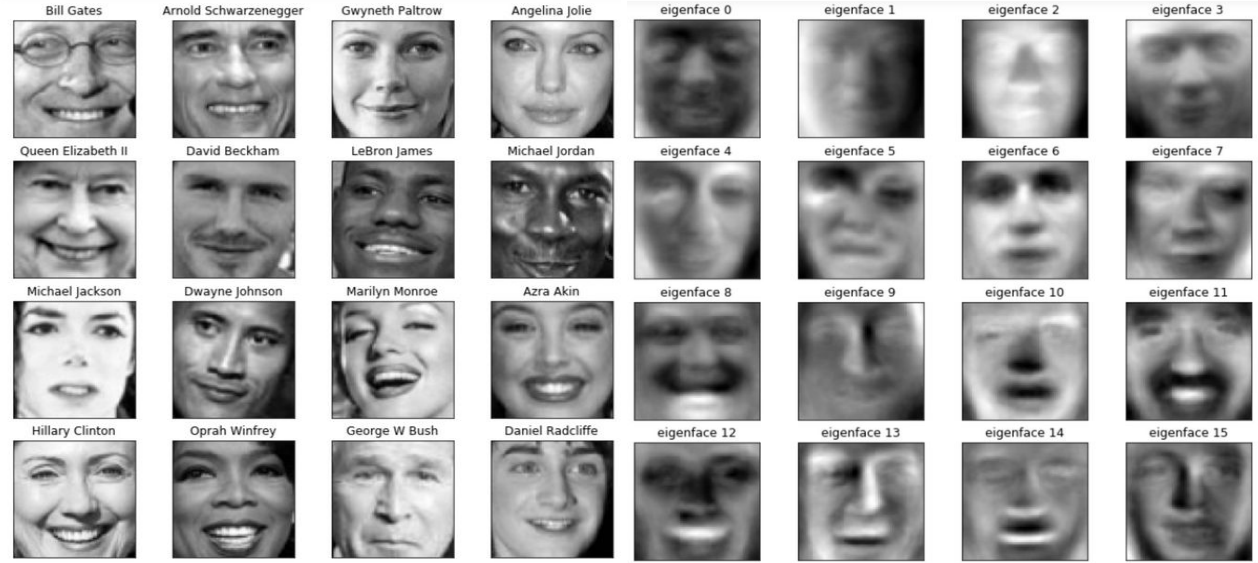
$$E_m = \frac{\|GU^\intercal - F\|^2}{\|F\|^2} = \frac{\lambda_{m+1} + \cdots + \lambda_n}{\lambda_1 + \cdots + \lambda_n} \leqslant \varepsilon.$$

Dimensionality reduction



Let’s walk through space…

Dimensionality reduction



16 components

Dimensionality reduction



50 components

Dimensionality reduction



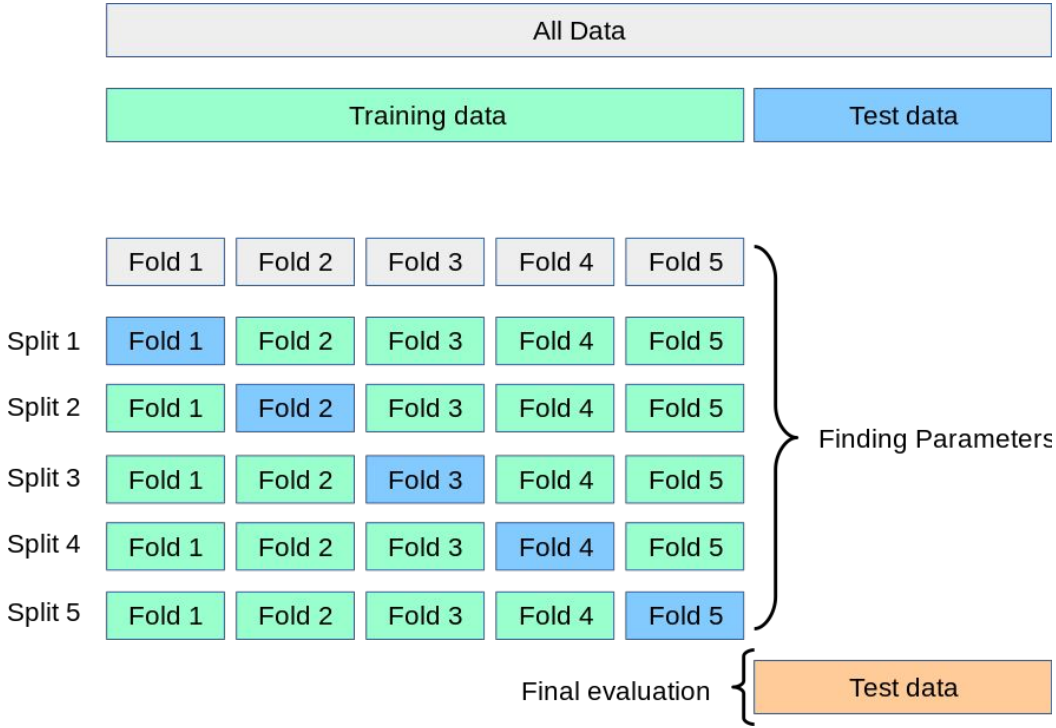
250 components

Validation strategies

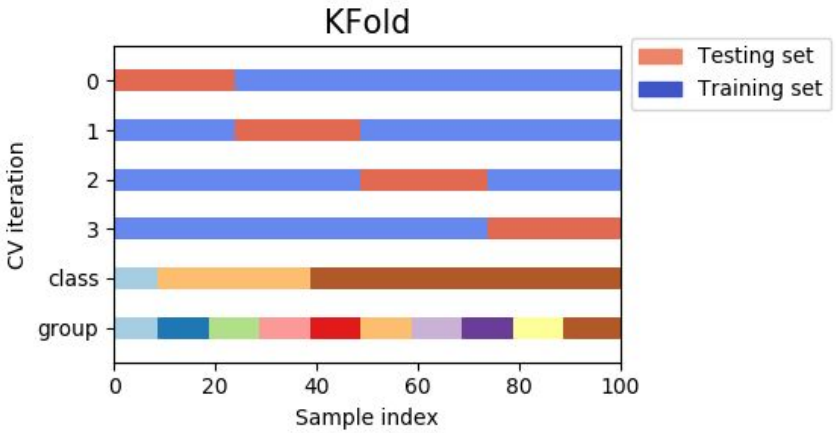
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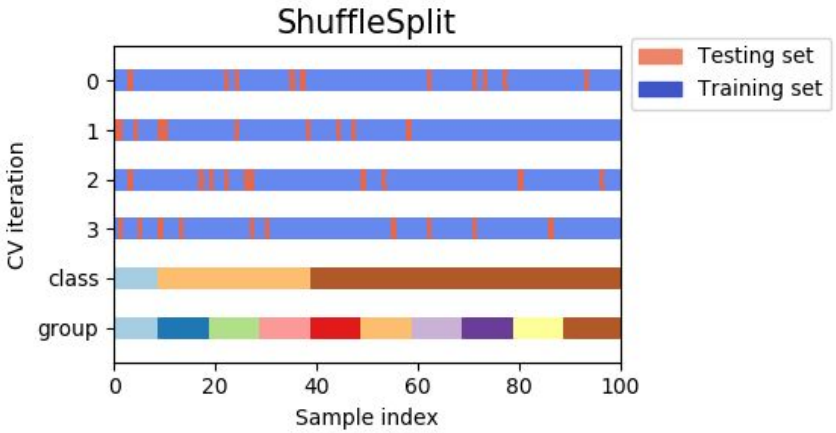
03

Validation strategies



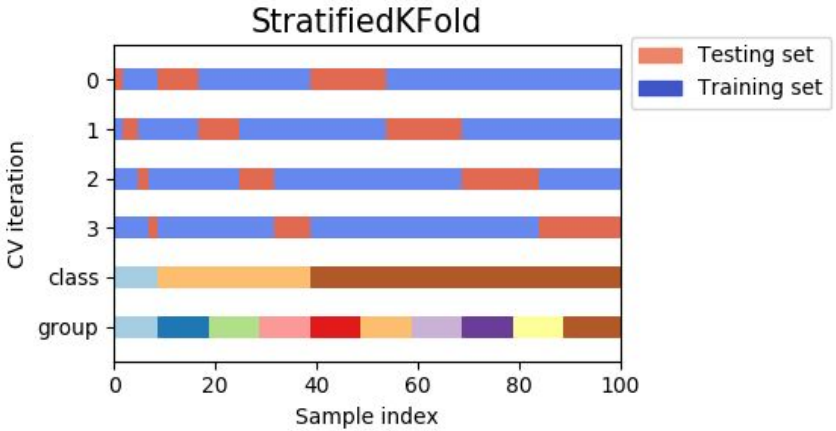
Validation strategies

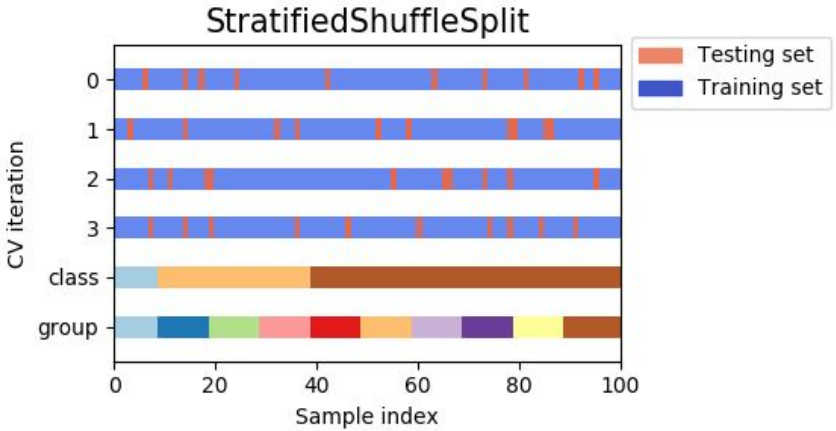




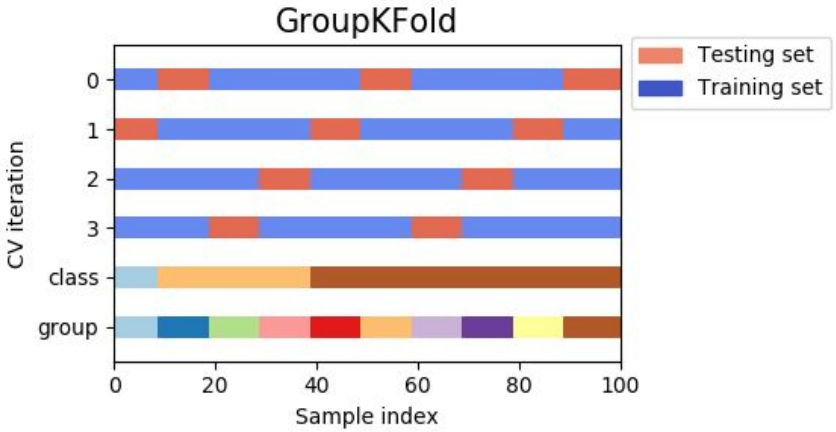
Special case: Leave One Out (LOO) - good for tiny datasets

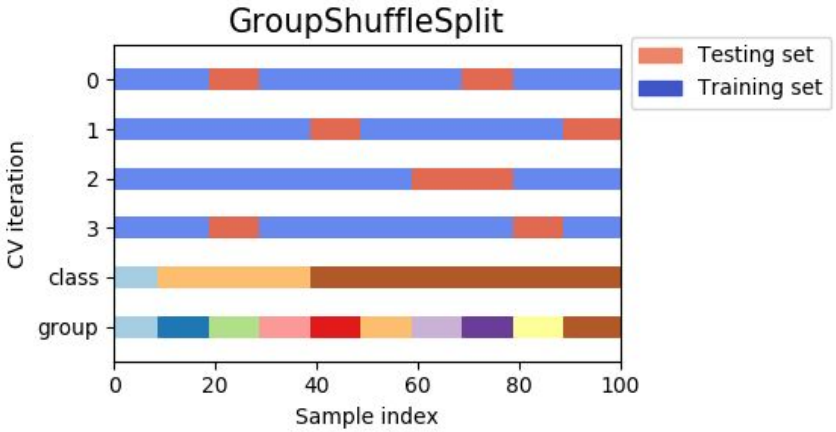
Validation strategies



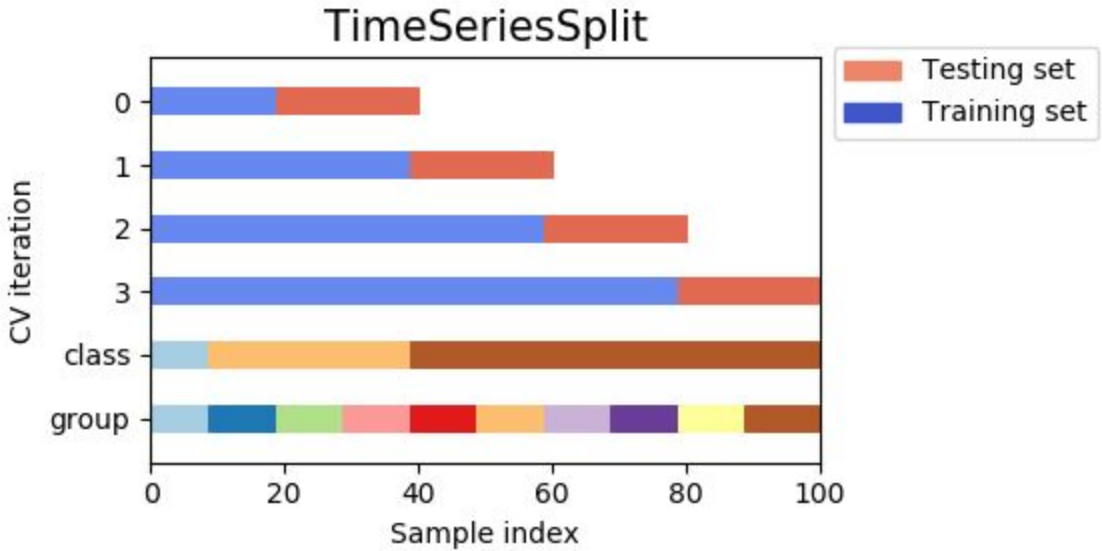


Validation strategies





Special case: time series



Never use train_test_split in this case!!!

Revise



1. Support Vector Machine (SVM)
2. Dimensionality reduction and PCA
3. Validation strategies

Thanks for attention!

Questions?



